

METHODOLOGY

HOW THE TALLY IS BUILT, START TO FINISH

This is a **weighted fan-poll aggregator with bootstrap uncertainty** — not an official vote count and not a black box. Every published number is derived from the formulas below, and every formula is derivable by hand from the poll log.

1 · THE OBSERVATION MATRIX

Each poll snapshot becomes **one row** in a long-format table. A row records the source, its grade, its sample size, the week, and one value per housemate — and what that value means depends on what the source reported:

OBS_TYPE	WHAT THE SOURCE REPORTED	STORED VALUE
full_share	a poll with each housemate's share	a number in [0, 1]
top_N / bottom_N	an official ranking naming leaders / bottom housemates	0 / 1 flag
rank	an ordered list with no shares	an integer rank

MATRIX

$M = \text{one row per } (source, week)$

IN PLAIN WORDS a spreadsheet where each line says “this source said this about these housemates this week”.

WHY keeping raw rows instead of one blended number means every figure can be traced to the poll that produced it, and a bad source can be down-weighted without deleting it. Rows are deduplicated latest-wins per (source, week), so a re-run can never double-count the same poll.

2 · RECENCY WINDOW AND WEIGHTS

Only the last **3 weeks** enter. Each row gets a weight from three things: how trustworthy its source is (q), how many votes it saw (n), and how old it is ($\lambda^{\Delta t}$).

ROW WEIGHT

$w_k = q_k \cdot \min(n_k, N_{cap}) \cdot \lambda^{\Delta t_k}$

SYMBOL	MEANING	VALUE NOW
q_k	source grade — trustworthiness	widget =1.0, transcribed =1.0, FB group =0.7, ranking channel =0.5
n_k	votes the poll saw	real votes, or a pseudo-n when hidden
N_{cap}	sample-size ceiling	5 000
λ	per-week decay	0.6
Δt_k	weeks between the row and now	0, 1, 2 ...
ϵ	constraint margin	0.01
B	bootstrap replicates	1 000

IN PLAIN WORDS a poll with far more votes counts more; a trusted source counts more; last week counts 60% of this week, and the week before counts 36%.

WHY the cap stops one viral widget (hundreds of thousands of votes) from drowning every other source; the decay keeps the snapshot responsive without throwing history away.

3 · SHARES ARE RENORMALISED INSIDE EACH POLL

Rarely do two polls list the same housemates. Each row is rescaled so its covered shares sum to 1, making sources comparable:

ROW SHARE

$$\tilde{s}_{i,k} = \frac{s_{i,k}}{\sum_j s_{j,k}}$$

IN PLAIN WORDS if a poll only listed six of nine housemates, we rescale those six to add to 100% before combining.

WHY without this, a source covering fewer people would look like it gave everyone a smaller share.

4 · AGGREGATION — THE WEIGHTED BLENDED SHARE

AGGREGATE SHARE

$$S_i = \frac{\sum_k w_k \cdot \tilde{s}_{i,k}}{\sum_k w_k}, \text{ full-share rows only}$$

IN PLAIN WORDS a weighted average of every poll, where bigger and more recent polls pull the result further.

WHY this is the published headline. It is a plain weighted mean — no softmax, no hidden function — so it can be checked with a calculator and the poll log.

5 · COVERAGE — NEVER INVENT A NUMBER

If no full-share source covered a housemate this week, they inherit last week's blended share (flagged carried); a housemate never covered at all is floored at the week's minimum measured share (flagged unmeasured).

CARRY-FORWARD

$$S_i \leftarrow S_i^{(x-1)} \text{ when } S_i = 0$$

IN PLAIN WORDS if nobody polled a housemate, we carry their last known value forward rather than pretending it is zero.

WHY a silent zero would look like “no support” when it is really “no data”. Carried values are labelled everywhere they appear.

6 · CONSTRAINTS — OFFICIAL RANKINGS, SOFTENED ON CONFLICT

When official bottom-N / top-N rankings exist, they bound the shares. If a constraint agrees with the polls it applies fully; if it contradicts them, the affected shares are pulled only **halfway** to the bound, and the softening is logged.

BOUNDS

$$\text{bottom-N: } S_i \leftarrow \min(S_i, m_{\text{out}} \cdot (1-\epsilon)) \cdot \text{top-N: } S_i \leftarrow \max(S_i, m_{\text{out}} \cdot (1+\epsilon))$$

ON CONFLICT

$$S_i \leftarrow S_i + \frac{1}{2} \cdot (\text{bound} - S_i)$$

IN PLAIN WORDS an official list is strong evidence, but one thin ranking report should not completely overrule thousands of votes.

WHY halfway pulls resolve disagreement safely instead of letting a single lower-grade row flip the standings; every adjustment is written to the run log.

7 · RENORMALISE, THEN BOOTSTRAP THE UNCERTAINTY

After carry and constraints the shares are renormalised to sum to 1. Then the whole weighting is repeated $B=1\,000$ times, each time resampling the poll rows in proportion to their weights (and drawing the polls' voters), giving the uncertainty band:

RENORMALISE

$$S_i \leftarrow \frac{S_i}{\sum_j S_j}$$

89% INTERVAL

[S_i (5.5) , S_i (94.5)] from B resamples

IN PLAIN WORDS we re-run the whole calculation a thousand times with slightly different poll samples; the range that contains the middle 89% is our uncertainty band.

WHY poll data is thin and noisy; an interval is more honest than a single over-confident number. Constraints are rules, not votes, so every replicate applies them — which keeps the point estimate inside its own band.

8 • FROM SHARES TO THE PUBLISHED PRODUCTS

Everything else is read straight off those same 1 000 replicates — no extra machinery:

WIN PROBABILITY

$$P(\text{win})_i = S_i$$

RANK CHIP

$$P(\#1)_i = \frac{1}{B} \sum_b \mathbf{1}\{\text{rank}_b(i) = 1\}$$

HEAD-TO-HEAD

$$P(A > B) = \frac{1}{B} \sum_b \mathbf{1}\{S_A(b) > S_B(b)\}$$

MOMENTUM

$$\Delta S_i = S_i(x) - S_i(x-1)$$

IN PLAIN WORDS “share = chance of winning”; a rank chip is just the fraction of the thousand runs in which that housemate finished in that position.

READING A MATCHUP $P(A > B)$ above 0.90 is a clear lead, 0.60–0.90 is a lean, and below 0.60 is “too close to call” (shown as a tie).

WHY using the same replicates for every product means the podium, chips and head-to-head can never disagree with one another.

9 • GAMBIT AUTO-FINALISTS

Housemates holding a guaranteed finale seat under the Gambit twist never enter the poll matrix (they are not in the save polls), so they carry no share and no win number. They are shown in their own strip for transparency.

WHY mixing ineligible housemates into win shares would inflate everyone's denominator and misstate the race.

10 • WHAT THIS IS — AND IS NOT

It is a transparent, reproducible aggregation of publicly available fan polls, with explicit uncertainty and a full audit trail.

It is not an official result, a vote count, or independent sampling. Fan polls share repeat-votable mechanics and recurring voters across sites, so treat the output as a **correlated fan-intensity index**. Numbers are shares — never votes. Every published surface carries the “fan forecast, not official result” disclaimer.